VIDEO DEEPFAKE DETECTION

Explainable AI Forensic Analysis Report

ANALYSIS ID
sample_000...
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FORENSIC VERDICT	MANIPULATION PROBABILITY	THRESHOLDS	CLIPS / DURATION
INCONCLUSIVE	**39.5%** P(REAL): 60.5%	Authentic: <0.30 Manipulated: ≥0.70	**30 clips** (12.0s) Time: 2.73s

Forensic Summary & Evidence Synthesis

Forensic analysis returned an INCONCLUSIVE verdict with P(FAKE) = 39.5%. The model's probability falls within the uncertain range (0.30 ≤ P < 0.70). Neither branch provided sufficient statistical confidence to confirm manipulation or verify authenticity. Potential factors include high compression, motion blur, or subtle blending.

Branch Attribution (Exact Shapley Decomposition)

Branch	Diagnostic Focus	Evidence Share (%)	Shapley Value (pl
Spatial Head (S)	Appearance, facial anatomy, synthesis boundary artifacts	13.8%	-0.0145
Temporal Head (T)	Motion continuity, temporal jitter, inter-frame coherence	86.2%	-0.0906

Facial Component Attribution (FCG Query Space)

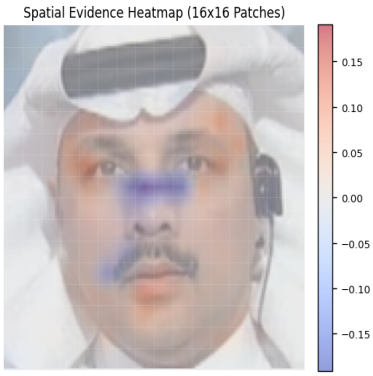
Facial Component	Signed Logit Contribution	Normalized Share (%)	Evidence Pushes Towards	Importance
Lips	+0.0737	7.5%	FAKE	LOW
Skin	+0.0628	6.4%	FAKE	LOW
Eyes	+0.2227	22.7%	FAKE	MEDIUM
Nose	-0.6224	63.4%	REAL	HIGH

Visual Forensic Heatmaps (16x16 Patch Decomposition)

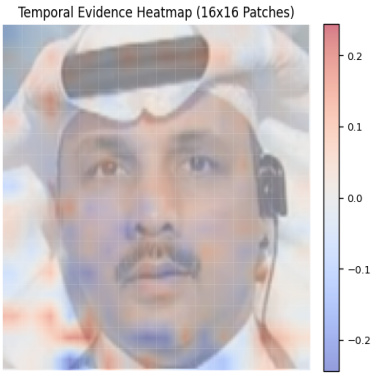
Extracted Face Crop



Spatial 16x16 Heatmap



Temporal 16x16 Heatmap



Video Quality & Robustness Assessment

Resolution: 640x480 | **FPS:** 25.0 | **Sharpness:** Crisp (211.07 var) | **Compression:** Moderate

- High image sharpness (Laplacian variance 211.1). Fine facial textures preserved.
- Balanced lighting (mean luminance 157.8/255).
- Standard definition video (640x480).

Methodology & Architecture: Yue-Hua Han et al., 'Towards More General Video-based Deepfake Detection through Facial Component Guided Adaptation for Foundation Model' (CVPR 2025, arXiv:2404.05583). Attributions computed via exact Shapley decomposition and additive LayerNorm-Linear projections. Formal video deepfake forensic analysis report.